

Mandatory Operating Systems Lab Report

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1 Linux: Reverse

1.1 Object

You will write a simple program called reverse. This program should be invoked in one of the following ways:

```
1 prompt> ./reverse
2 prompt> ./reverse input.txt
3 prompt> ./reverse input.txt output.txt
```

The goal of the reversing program is to read in the data from the specified input file and reverse it; thus, the lines should be printed out in the reverse order of the input stream.

1.2 Experimental Procedure

- 1 Write the reverse.c file
- 2 Compile it
- 3 Run/test it

1.3 Results

All test cases passed.

2 Xv6: Xv6 Syscall

2.1 Object

The goal of the project is simple: to add a system call to xv6. Your system call, `getreadcount()`, simply returns how many times that the `read()` system call has been called by user processes since the time that the kernel was booted.

Your new system call should look have the following return codes and parameters:

```
int getreadcount(void)
```

Your system call returns the value of a counter (perhaps called `readcount` or something like that) which is incremented every time any process calls the `read()` system call. That's it!

2.2 Experimental Procedure

- 1 Modify the sysfile.c, syscall.h, syscall.c, usys.S and user.h files.
- 2 Compile it
- 3 Run/test it

2.3 Results

All test cases passed.